



ROBERTO NOGUEIRA

AI-Native Engineer | Ruby on Rails | React | Agentic Systems
Software & Telecom Engineer | AI-Native Product Architect

📍 São José dos Campos, SP, Brazil



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PROFESSIONAL SUMMARY

AI-Native Software **Engineer** with a strong Telecom background and **7+ years** of **Ruby on Rails** experience. Skilled in **Ruby, React, JavaScript, Python**, and **Elixir**, building agentic systems, AI-native knowledge systems, modular monoliths, microservices, scalable backend architectures, CI/CD, automation, and workflow integration. Deep experience in **OSS/BSS**, multinational system integration, **billing, payroll, subscriptions, API integrations**, mentoring, technical leadership, and reliable software delivery at scale.



VALUE HIGHLIGHTS

- ✓ **7+ years** of experience with **Ruby on Rails** and strong backend / full-stack delivery.
- ✓ **International engineering experience** across LATAM, USA, Canada, and Europe.
- ✓ **300%** productivity improvement through Ruby/Rake automation at Ericsson.
- ✓ Experience with **payroll, billing, subscriptions**, event-driven integrations, and async processing.
- ✓ **AI-assisted SDLC, agentic workflows, multi-agent systems**, and context engineering.

AI-NATIVE PRODUCT ARCHITECTURE

FROM INTENT AND KNOWLEDGE TO REAL-WORLD IMPACT



COSMOS

AI-native operating system connecting knowledge, meaning, intelligence, intent, and creation.



HOLOFLUX

Semantic runtime and visual language for intent, inquiry, evidence, execution, and understanding.



IOP

Experimental AI-native programming model for agentic systems and next-generation software development.



FACTORIES

Engines of explication that transform structured meaning and intent into communicable digital media.



CORE COMPETENCIES



Languages:

Ruby, Python, JavaScript, Elixir, Erlang/OTP, Java



Frameworks & Frontend:

Ruby on Rails, React, Phoenix, Vue, jQuery, Bootstrap



AI & Agentic Engineering:

AI-Native Software Engineering, AI-Assisted SDLC, Agentic Workflows, Multi-Agent Systems, Context Engineering, GitHub Copilot, OpenAI Codex, Claude Code, Warp



Architecture:

Modular Monoliths, Packwerk, Microservices, REST, GraphQL, Clean Architecture, DDD, Event-Driven Architecture, Kafka



Backend & Async Processing:

Sidekiq, Redis, Background Jobs, API Integration



Databases:

PostgreSQL, MySQL, Oracle/PLSQL, MongoDB, Neo4j, Sybase



Testing & Quality:

RSpec, Minitest, PyTest, JUnit, ExUnit, SimpleCov, RubyCritic, Skunk



Security:

Brakeman, bundler-audit, bundler-leak



DevOps & CI/CD:

Git, GitHub, GitHub Actions, GitLab, Bitbucket, Docker, Kubernetes, Terraform, Helm, Puppet



Cloud & Platforms:

AWS, Heroku, Engine Yard, OpenStack, StarlingX, IBM Cloud



Observability & Tools:

Datadog, Kibana, Redash, Rollbar, OpenAPI/Swagger, Postman



Telecom & OSS/BSS:

Ericsson OSS-RC, BSS, MSS, APG, AXE, CPP, MOShell



Engineering Practices:

Clean Code, SOLID, DRY, Design Patterns, Scrum

TURNING COMPLEXITY INTO UNDERSTANDING



PROFESSIONAL EXPERIENCE

1



06/2013 — Present

Zootworks — São José dos Campos, Brazil**Founder & Consultant | AI-Native Software Engineer**

Summary: Providing software engineering and technology consulting focused on AI-native development, Ruby on Rails, systems integration, telecom, and backend platforms.

Key achievements:

- Built multi-agent and AI-first frameworks for eBooks, career workflows, technical blogs, and automation.
- Built an event-driven Ruby on Rails microservice with Kafka to orchestrate student subscription workflows and external API integrations.
- Integrated AI-driven workflows into GitHub-based engineering processes.
- Used AI and code-quality tooling to automate Rails technical-debt analysis and modernization prioritization.
- Developed scalable backend services with Ruby, Elixir, and Erlang/OTP.



Projects: AI-native frameworks — COSMOS / HOLOFLUX / IOP / FACTORIES | Anubis/CX (Qeevo) | PicPay (IEBT) | Track Server (ITER) | Adaive (Lab360) | Taxweb Auditor (GSW) | OSS Support (Entel)

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06/2021 — 09/2024

Encora — Campinas, Brazil**Senior Ruby on Rails Engineer | System Specialist**

Summary: Worked on payroll, billing, and Employer Risk systems at Justworks, plus cloud/edge initiatives at Wind River.

Key achievements:

- Developed scalable payroll and billing workflows using Ruby on Rails, React, GraphQL, Redis, and Sidekiq.
- Worked on Workers' Comp billing including rates, policies, eligibility, error handling, fallback, and reconciliation.
- Built reliable asynchronous processing solutions with Sidekiq/Redis, including bulk S3 workflows.
- Maintained a Rackwerk modular monolith and modernized legacy interfaces with React.
- Used CI/CD and observability with GitHub Actions, Datadog, Kibana, Redash, and Rollbar.
- Technical leadership: interviewing, mentoring, and collaboration.



Projects: PEO / Payroll Platform | Time Tracker | WRCP (Wind River)

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07/2018 — 06/2021

InMov — São Paulo, Brazil**Senior Ruby on Rails Engineer (Full Stack)**

Key achievements:

- Developed Smart City applications using Ruby on Rails, JavaScript, jQuery, and AJAX.
- Led Ruby on Rails upgrades from version 4.2 to 6.0, improving performance, security, and maintainability.
- Created Site Rails Manager, a Bash/Docker CLI toolkit to automate and standardize Ruby on Rails development environments and workflows.



Projects: City Hall Permit System | Site Rails Manager

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12/2004 — 06/2013

Ericsson — Indaiatuba, SP, Brazil**Software Developer | Solution Integrator | Project Manager**

Key achievements:

- Developed, integrated, and deployed telecom systems in large-scale OSS/BSS environments.
- Led international OSS/BSS initiatives across LATAM, USA, Canada, and Europe.
- Built MOP Automation using Ruby and Rake, achieving 300% productivity improvement.
- Delivered technical training and knowledge transfer in international projects.



Projects: BSS Dual Stack | Full IP Rehomming | MOP Automation | Revenue Assurance (Vivo) | OSS/BSS Upgrades & Support



EDUCATION

1998 — Data Communications — USP

1994 — Software Engineering (Postgraduate) — UNIAP

1985 — Telecommunications / Electronics Engineering (Bachelor's Degree) — INATEL



CERTIFICATIONS

SAP Integration — Thomson Reuters

Solution Integrator Experienced (Job Stage 5) — Ericsson

Getting and Cleaning Data — Johns Hopkins University

Functional Programming in Erlang — Kent University

R Programming — Johns Hopkins University



COURSES & SPECIALIZATIONS

2026 — Claude Code: The Complete Course — Udemy

2026 — GitHub Copilot CLI — GitHub

2025 — AI Automation — Udemy

2025 — MCP Crash Course — Udemy

2025 — Working with Vue 3 and Go — Udemy

2025 — Working with Microservices in Go — Udemy

2025 — GitHub Copilot — Udemy

2023 — Metaverse — Coursera

2019 — Elixir & OTP — The Pragmatic Studio

2019 — Elixir: Scalable and Efficient Application Development — Udemy

2017 — Specialization in Erlang Programming — Kent University / FutureLearn

2016 — Specialization in Ruby on Rails Web Development — Johns Hopkins University / Coursera

2015 — Web Applications Architectures — University of New Mexico / Coursera

2014 — Specialization in Data Science — Johns Hopkins University / Coursera

2010 — Ruby on Rails Advanced — Caelum

2010 — Ruby on Rails — Caelum



LANGUAGES

Portuguese — Native / Bilingual

English — Full Professional

Spanish — Intermediate